Question:
John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for $15 each. He ends up buying 8 video games for $20 each and has $5 left. How many lego sets does he still have?

Reasoning + Answer:

Let's solve this step by step.

John plans to sell all his toys (specifically, Lego sets) and use the money to buy video games. He sells 13 Lego sets at $15 each.

Step 1: Calculate total money from selling Lego sets.

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13 \text{ sets } \times \$15 = \$195
$$

Step 2: He buys 8 video games at $20 each.

$$
8 \text{ games } \times \$20 = \$160
$$

Step 3: He has $5 left after buying the games.

So, total money spent = money from Lego sales - money left

$$
\text{Total spent} = \$195 - \$5 = \$190
$$

But he spent $160 on video games, so:

$$
\text{Money used} = \$160
$$

$$
\text{Money left} = \$5
$$

So, total money earned = $160 + $5 = $165

Wait âĢĶ this contradicts the earlier calculation. Let's check again.

He sells 13 Lego sets at $15 each âĨĴ total money = 13 ÃĹ 15 = $195

He spends $20 ÃĹ 8 = $160 on video games

He has $5 left.

So total money spent = $160 + $5 = $165

But he earned $195 from selling Legos.

So the difference is:

$$
195 - 165 = \$30
$$

So he had $30 left over from the sale, but he only has $5 left. This suggests a contradiction unless the question is asking something else.

Wait âĢĶ the question says:

"John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for $15 each. He ends up buying 8 video games for $20 each and has $5 left. How many lego sets does he still have?"

Wait âĢĶ if he sells **all** his toys, then he should sell all 13. But the question is asking: "How many lego sets does he still have?" That would only be relevant if he didn't sell all of them.

So the key point is: the initial statement says he plans to sell **all** his toys. But then it says he ends up buying 8 video games and has $5 left.

So perhaps he didn't sell all the Lego sets âĢĶ maybe he only sold some, and the leftover money is used to buy video games.

But the question says he has 13 Lego sets and sells them for $15 each âĢĶ does that mean he sells all of them?

Yes, the sentence: "He has 13 lego sets and he sells them for $15 each" âĢĶ this suggests he sells all 13.

So total money from sale = 13 ÃĹ 15 = $195

Then he buys 8 video games at $20 each = $160

Money left = $5

So total money spent = 160 + 5 = $165

But he earned $195, so money left should be 195 - 165 = $30

But the problem says he has $5 left.

So contradiction.

Therefore, the only logical conclusion is that he did **not** sell all 13 Lego sets.

So the question must be: He has 13 Lego sets. He sells some of them for $15 each. Then uses the money to buy 8 video games at $20 each, and has $5 left. How many Lego sets does he still have?

So he sells some, not all.

Let x be the number of Lego sets he